



## DO-IT-YOURSELF: CIRCUIT IDEA

adjusted using the inbuilt potentiometer (blue) on the module. The sensor probe should be inserted a few centimetres into the soil.

The soil moisture module's output provides a low or high signal based on soil dryness or wetness. When the soil is wet, the digital output (D0) of the sensor is high, preventing the PNP transistor (BC558, T1) from conducting. When the soil is dry, the digital output (D0) goes low, enabling T1 to conduct. The output of the soil sensor is fed to the base of T1 through resistor R1. When dryness is detected, T1 conducts, energising the relay (RL1) and switching on the water pump for irrigation.

RL1 energises and switches from normally closed (NC) to normally open (NO) based on soil dryness. The NO and common contacts are used to control the water pump, managing garden irrigation.

### Decorative lighting

The decorative lighting system consists of eight different coloured LEDs, an NPN transistor (BD139, T2), an on/off switch (S1), and various other components. The power supply is common to both the systems. The switch (S1) turns the decorative lights on and off.

The LEDs should be arranged at various points in the garden, utilising twin wires. When powered through switch S1, the LEDs light up, enhancing the garden's decorative appearance. During the daytime, the lights can be turned off using S1.

### Construction and testing

An actual-size, single-sided PCB layout for the circuit is shown in Fig. 3, with the component layout depicted in Fig. 4. After assembling the circuit on the PCB, it should be enclosed in a suitable low-cost PVC box.

Connectors (CON1 and CON2) should be attached to the front for connections to the 230V AC mains and the water motor, respectively. Switch S1 should be mounted on the front to control the decorative lights. The LEDs should be extended with twin wires for proper garden decoration. **EFY**

**Bonus.** You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/make-decorative-automated-garden-watering>

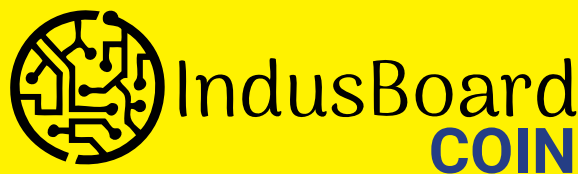
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